The Delivery Is Not the Observation: Effective Sample Size in Ball-by-Ball Win Probability Models for Twenty20 Cricket

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***Abstract*—Ball-by-ball models in cricket are fitted on hundreds of thousands of deliveries but only around a thousand independent outcomes, because every delivery of a match shares a single label. We show that ignoring this gap corrupts four distinct stages of a modelling pipeline, using a calibrated win-probability model over 286,621 Indian Premier League deliveries from 1,195 matches (2007/08–2026) as the instrument. First, *model selection*: a conventionally tuned gradient-boosted ensemble attains the *best* validation loss and the *worst* test loss, generalising 20 % worse than a heavily regularised one, and validation loss is flat across the entire leaf-size range over which test loss moves—so the failure is invisible to standard tuning. Second, *calibration*: isotonic regression fitted on one held-out season, 71 independent outcomes, worsens expected calibration error from 0.032 to 0.054. Third, *rating stability*: Win Probability Added is three to four times less persistent season-over-season than strike rate, because 10.0 % of deliveries carry 17.9 % of all win-probability movement. Fourth, *validation power*: with 140 test matches a squad-rating design cannot resolve effects below 0.09 AUC. A team-level decomposition that is exact by construction sharpens the point: for the most-capped franchise in the league, 259 matches across nineteen seasons still cannot resolve whether its margin is built with bat or ball (difference $+0.030$, 95 % CI $[-0.026, +0.084]$). We recommend that regularisation and validation for such models be specified in independent matches rather than deliveries. The mechanism is not cricket-specific, and we argue it should be expected wherever labels are shared within a contest.**

***Index Terms*—effective sample size, clustered data, model selection, cross-validation, model calibration, win probability, sports analytics, Twenty20 cricket**

I. Introduction

A Twenty20 cricket match is approximately 240 deliveries long, and a season of the Indian Premier League produces roughly 17,000 of them. A modelling pipeline that predicts the eventual winner from the state before each delivery therefore appears to have an abundant dataset: our corpus contains 286,621 rows. It does not. The label—which team won — is constant across every delivery of a match. The corpus contains 1,195 independent outcomes, not 286,621, and the ratio between those two numbers is roughly 240.

This is a familiar phenomenon in statistics under several names: the design effect in survey sampling [1], intra-cluster correlation in hierarchical models, and the motivation for grouped resampling schemes [2]. It is also, in our experience, routinely overlooked in applied sports-analytics pipelines, where the delivery, pitch, or possession is treated as the unit of observation because it is the unit of the data file. The consequences are not confined to standard errors being too small. They reach into model selection, calibration, the interpretation of derived ratings, and the design of validation experiments—and in the first of those, the failure is silent.

This paper documents that failure and its consequences. We build a calibrated ball-by-ball win-probability model for Twenty20 cricket and the Win Probability Added (WPA) rating derived from it, then use that pipeline as an instrument to demonstrate four distinct ways in which treating deliveries as observations misleads the analyst.

The instrument itself is not new in concept. Win expectancy models originate with Lindsey's work on baseball [4], and WPA has been a standard currency of contribution in that sport for decades [5], with analogues in American football [6], basketball [7], and association football [8]. Cricket has a mature literature on resource and outcome modelling [9]–[11]. Our contribution is not the metric but what building it carefully reveals.

1) **A silent model-selection failure.** An unregularised gradient-boosted ensemble achieves the best validation loss and among the worst test loss. Validation loss varies by less than 0.02 nats across a leaf-size range over which test loss moves 0.10 nats, so a practitioner tuning in the standard way receives no warning. We show that regularisation must be expressed in independent matches, and that a leaf holding 30 deliveries holds roughly one eighth of one observation.

2) **Three further consequences** of the same cause, each quantified: post-hoc calibration fitted on a single season degrades calibration; a leverage-concentrated rating is far less persistent than the aggregates it aims to replace; and a validation design with 140 test matches is underpowered for the effects it is asked to detect.
3) **The win-probability instrument**, described in enough detail to reproduce: an era-normalised state representation that survives two decades of scoring inflation, a delivery-level attribution scheme with explicit handling of the innings boundary, extras and dismissals, and a team-level aggregation that decomposes a match result exactly into batting and bowling contributions.
4) **A worked demonstration** on the full IPL record, including a head-to-head decomposition for its most-capped franchise which shows both what aggregation buys and where it stops: the exact identity holds, the aggregate margin is marginally resolvable, and the batting/bowling split is not resolvable at all even over 259 matches.

Section II reviews related work. Section III describes the data and the state representation. Section IV presents the model, the attribution scheme and the team-level identity. Section V details the protocol. Section VI reports the four consequences and the demonstration, and Section VII discusses implications and limitations.

II. Related Work

A. Clustered Data and Effective Sample Size

That correlated observations carry less information than independent ones is long established. Kish's design effect [1] quantifies the inflation of sampling variance under clustering, and implies an effective sample size below the nominal count. Field and Welsh [2] analyse bootstrap schemes for clustered data and show that resampling must respect the cluster structure to be consistent. In predictive modelling the corresponding concern is that cross-validation folds sharing structure with the training set produce optimistic estimates; Roberts et al. [3] survey blocking strategies for temporally, spatially and hierarchically structured data and argue that validation design must mirror the dependence structure.

Our contribution relative to this literature is not the statistical principle, which is standard, but its consequences in a setting where the clustering is extreme—roughly 240 correlated rows per independent label—and where, as we show, the resulting model-selection failure is invisible to the validation procedure that a practitioner would naturally run.

B. Resource and Outcome Models in Cricket

The Duckworth–Lewis method [9] models a batting innings as the consumption of two resources—deliveries remaining and wickets remaining—and provides a resource table permitting fair target revision after interruption. Stern [10] recalibrated the method for the elevated scoring rates of the Twenty20 era. These models are outcome-neutral: they estimate expected scoring capacity, not victory probability. Bailey and Clarke [11] demonstrated in-progress outcome prediction for one-day internationals using regression on match state. Davis, Perera, and Swartz [12] developed a full ball-by-ball simulator for Twenty20, which supports counterfactual analysis but is computationally heavier than a direct discriminative model. Within the IPL specifically, Kumar et al. [20] and Vistro et al. [21] apply supervised learning to outcome prediction. None of this work reports effective sample size, and all of it validates at the delivery or match-state level.

C. Player Valuation Metrics

Cricket-specific composite metrics have been proposed to address the deficiencies of raw averages—for example Lemmer's combined bowling rate [13], which blends average, economy, and strike rate into a single index. Sharma [23] likewise reduces Twenty20 performance to a small set of factors extracted from aggregate match statistics. Such measures remain aggregate and context-blind: they reweight summary statistics rather than conditioning on match state. Saikia et al. [22] model IPL auction valuation from performance history, again without conditioning on situation. In other sports, the dominant modern approach values individual *actions* by the change in expected outcome they induce: VAEP [8] for soccer, nflWAR [6] for American football, and expected possession value [7] for basketball. The instrument we build is the cricket analogue of that family.

D. Probability Calibration and Interpretability

Because the derived rating is a *difference* of two predicted probabilities, calibration matters at least as much as discrimination: a model that ranks states correctly but reports compressed probabilities produces compressed ratings. We evaluate with proper scoring rules [16] and examine post-hoc calibration [17], [18]—which, as Section VI-C shows, is itself subject to the effective-sample-size problem. Model interpretability is provided by SHAP [15], whose tree estimator is exact for the gradient-boosted ensembles [14] used here.

III. Data and Preprocessing

A. Source

We use ball-by-ball records for the Indian Premier League obtained from Cricsheet [19], which publishes structured per-delivery data for every match, including batter, non-striker, bowler, runs scored, extras by type, dismissal type and player dismissed, and match-level metadata (venue, toss, result, announced playing XI, and Duckworth–Lewis application). The raw corpus comprises 1,243 matches and 295,732 delivery records spanning seasons 2007/08–2026.

B. Filtering

We exclude matches with no result and matches abandoned before the completion of the first innings, since no outcome label exists (9 matches). Matches decided by the Duckworth–Lewis–Stern method (23) and matches tied and decided by Super Over (16) are also excluded: in the former the effective target changes mid-innings, so the state representation and the label refer to different contests, and in the latter the result is

determined outside the 40-over structure the model describes. Super Over innings themselves are dropped wherever they appear. This leaves 1,195 matches and 286,621 deliveries. Table I summarises the retained corpus. The ratio of the last two columns—roughly 240—is the quantity this paper is about.

TABLE I
DATASET COMPOSITION AFTER FILTERING

Split	Seasons	Matches	Deliveries
Train	2007/08–2023	984	236,047
Validation	2024	71	17,103
Test	2025–2026	140	33,471
Total	2007/08–2026	1,195	286,621

C. Match State Representation

For each delivery i we construct a feature vector describing the state of the match *immediately before* the ball is bowled, ensuring that no information from the delivery's own outcome enters its own prediction:

$$s_i = \big(\iota,\ b_i,\ w_i,\ R_i,\ \rho_i,\ \rho_i^*,\ T,\ \delta_i,\ \phi_i,\ \nu_i,\ \tilde{R}_i,\ \tilde{\rho}_i^*,\ \tilde{T}_i\big)$$

where $\iota \in \{1, 2\}$ is the innings index; b_i is deliveries remaining in the innings; w_i is wickets remaining; R_i is the batting team's current score; $\rho_i = 6R_i/(120-b_i)$ is the current run rate; $\rho_i^* = 6(T - R_i)/b_i$ is the required run rate (second innings only); T is the target; $\delta_i = T-R_i$ is runs still required; and ϕ_i is the innings phase (powerplay, overs 1–6; middle, 7–15; death, 16–20).

The remaining four features are era-relative and are essential rather than cosmetic. ν_i is the *venue par score*: the mean first-innings total at that venue over the three immediately preceding seasons, shrunk toward the contemporaneous league mean with a 20-match prior, and computed using only seasons strictly preceding the current match so that no look-ahead leakage occurs. The final three features normalise the absolute quantities by that par score: $\tilde{R}_i = R_i/\nu_i$, $\tilde{T}_i = T/\nu_i$, and $\tilde{\rho}_i^* = \rho_i^*/(\nu_i/20)$.

We initially used an all-time expanding venue mean and absolute run totals alone, as is conventional. That specification fails badly out of sample. Scoring rates in the IPL have inflated sharply over the study period, so a model keyed to absolute totals reads an inflated first-innings score as dominant when it is merely par: on the test seasons it predicted a mean win probability of 0.579 for first-innings states whose realised win rate was 0.424. A rolling par score and par-relative features let the ensemble generalise across eras instead of memorising absolute thresholds, and we recommend the same treatment for any Twenty20 model spanning more than a few seasons.

Features undefined in the first innings (ρ_i^*, T, δ_i, $\tilde{T}_i$, $\tilde{\rho}_i^*$) are encoded as missing and handled natively by the tree ensemble's default-direction mechanism [14].

D. Label

The label $y \in \{0, 1\}$ indicates whether the team batting during delivery i went on to win the match. It is constant across every delivery of a match. Everything in Section VI follows from that sentence.

IV. THE WIN PROBABILITY INSTRUMENT

A. Model

We model

$$\hat{p}_i = \Pr\big(\text{batting team wins} \mid s_i\big)$$

using a gradient-boosted decision tree ensemble [14] trained with logistic loss. The ensemble is chosen not because it is the most accurate specification available—Section VI-A shows it is not—but because the attribution and interaction analyses that follow require a model class that represents feature interactions explicitly and admits an exact tree-based SHAP estimator. A single model is fitted across both innings, with the innings indicator ι and the missing-value mechanism allowing it to learn regime-specific structure.

Hyperparameters are selected on the validation season by grid search over `n_estimators` $\in \{300, 600\}$, `max_depth` $\in \{3, 4\}$ and `min_child_weight` $\in \{500, 2000, 5000\}$ at a fixed learning rate of 0.05, row and column subsampling of 0.8, and L_2 penalty 5.0: twelve configurations in all. The minimum-leaf grid is floored at 500 deliveries. Section VI-B explains why that floor is necessary and what happens without it; we state here only that the floor is a consequence of the paper's central finding rather than an arbitrary choice. The selected configuration is reported in Table II.

TABLE II
SELECTED MODEL HYPERPARAMETERS

Parameter	Value
Boosting rounds	600
Maximum tree depth	4
Learning rate	0.05
Min. leaf size (deliveries)	2000
Row subsample	0.8
Column subsample	0.8
L_2 regularisation	5.0
Calibration	Isotonic (pooled)

B. Calibration as a Designed Comparison

Because the derived rating is a difference of probabilities, miscalibration propagates directly into it. We therefore treat calibration as an experimental variable rather than a fixed processing step, and fit isotonic regression [18] two ways.

Variant A fits the calibrator on the validation season alone. This is the textbook recipe, and it supplies 71 independent outcomes. *Variant B* pools out-of-fold predictions across the training seasons—generated by a four-fold split grouped by match—with the validation-season predictions, supplying roughly 1,000. No test-season information reaches the

calibrator in either variant. The comparison is reported in Section VI-C and is one of the four consequences this paper documents.

C. Baselines

1) Logistic regression: A regularised logistic regression fitted on the identical feature set, with continuous features standardised, missing values imputed by a sentinel, and phase one-hot encoded. This isolates the contribution of non-linearity and feature interaction.

2) Resource-based baseline: Following the resource formulation of Duckworth–Lewis [9], [10], we estimate the expected remaining runs $\mathcal{R}(b_i, w_i)$ empirically from first-innings training data, project a final score $\hat{S}_i = R_i + \mathcal{R}(b_i, w_i)$, and map the projected margin $\hat{S}_i - T$ (second innings) or $\hat{S}_i - \nu_i$ (first innings, against the venue par score) to a win probability via a one-dimensional logistic fit estimated separately per innings. This is the natural domain-standard comparator.

D. Win Probability Added

Let $\hat{p}_i$ denote the calibrated win probability for the team batting at delivery i. The impact of delivery i is

$$\Delta_i = \hat{p}_{i+1} - \hat{p}_i,$$

and credit is assigned as $\mathrm{WPA}_i^{\mathrm{bat}} = +\Delta_i$ to the striker and $\mathrm{WPA}_i^{\mathrm{bowl}} = -\Delta_i$ to the bowler, since a gain in the batting team's win probability is by construction an equal loss for the bowling team. The non-striker receives no credit.

1) Innings boundary: At the final delivery of the first innings the batting team changes, so $\hat{p}$ is no longer defined on the same quantity. We restore continuity by defining the post-state of that delivery as the *complement* of the model evaluated at the opening chase state (ι=2, b=120, w=10, T=R_{final}+1), so that the last ball of the first innings is credited with the final increment of the target it sets. At the final delivery of the second innings the post-state is the realised outcome y, so the winning hit receives the full residual increment.

2) Extras and dismissals: Wides and no-balls are attributed wholly to the bowler with no batter credit. Byes and leg-byes are attributed to neither and recorded as unattributed residual, amounting to 1.07 % of total |WPA| mass. Bowler-credited dismissals (bowled, caught, caught and bowled, LBW, stumped, hit wicket) are attributed to the bowler. Run-outs, retirements out and obstruction carry no bowler causation; the full negative increment goes to the dismissed batter and the bowler credit is zero.

3) Aggregation: A player p's seasonal rating is $\mathrm{WPA}_{p,\sigma} = \sum_{i \in \mathcal{B}_{p,\sigma}} \mathrm{WPA}_i$ over the deliveries $\mathcal{B}_{p,\sigma}$ in which p appeared in the relevant role. Because raw WPA scales with playing time we also report $\mathrm{WPA}_{100} = 100 \cdot \mathrm{WPA}_{p,\sigma}/|\mathcal{B}_{p,\sigma}|$, with a qualification threshold of 120 deliveries.

4) An exact team-level identity: At team level no attribution question arises, because every increment belongs to one side or the other. Summing the raw increment over the deliveries a team batted, and its negation over those it bowled, gives

$$\mathrm{WPA}_{\mathrm{team}}^{\mathrm{bat}} + \mathrm{WPA}_{\mathrm{team}}^{\mathrm{bowl}} = y_{\mathrm{team}} - \hat{p}_0,$$

where $\hat{p}_0$ is the pre-match win probability, and the two sides of a match sum to zero identically. We verify this numerically: across all 1,195 matches the maximum absolute departure from zero is 0.0, and the mean team total is $+0.497$ for winners and -0.497 for losers. This identity is what makes Section VI-I well posed, and it is why that analysis uses the raw increment rather than the player-level series: the extras convention above deliberately withholds batter credit, which would bias a team-level split.

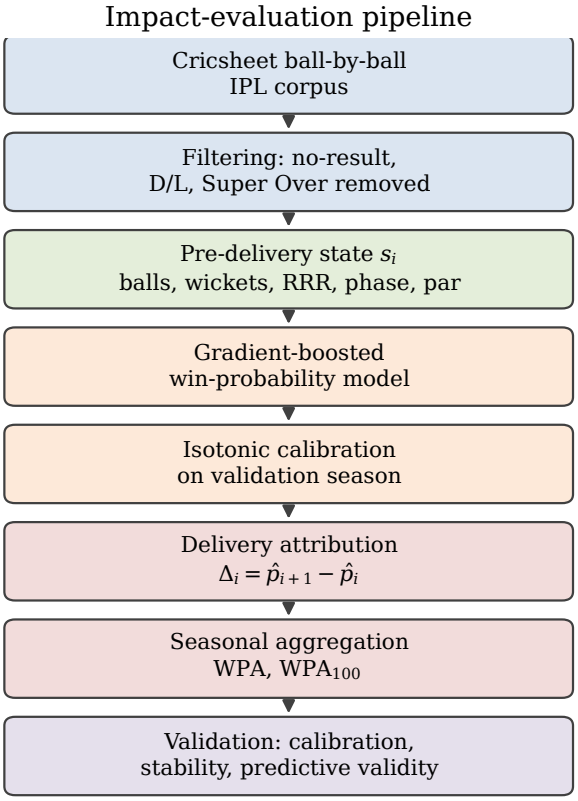


Fig. 1. End-to-end architecture of the win probability instrument.

V. Experimental Setup

A. Temporal Splitting

All splits are performed *at the match level and chronologically*. Splitting at the delivery level would place highly correlated rows on both sides of the partition and produce severely optimistic estimates. Seasons 2007/08–2023 form the training set, 2024 the validation set, and 2025–2026 the held-out test set. No test-season information is used in feature construction, hyperparameter selection, or calibration.

B. Effective Sample Size and Uncertainty

Although the corpus contains 286,621 deliveries, the effective number of independent outcome observations is approximately the number of matches, 1,195. Every confidence interval in this paper is obtained by bootstrap resampling *of*

matches, not of deliveries, with 500 replicates. We apply this without exception, including to the descriptive analysis of Section VI-I, where it materially changes what can be claimed.

C. Evaluation Metrics

Model quality is assessed with log loss, Brier score [16], AUC-ROC, and expected calibration error (ECE) over 10 equal-frequency bins, disaggregated by innings and phase. Rating quality is assessed by **temporal stability**—Spearman rank correlation between a player's rating in consecutive seasons, restricted to players qualifying in both, against the same statistic for the conventional aggregates—and **predictive validity**, in which squad-level ratings aggregated over the announced playing XI predict the following season's match winners.

D. Rating Basis

Held-out metrics come from the temporally split model. Player ratings must cover every season, so they are computed from *out-of-fold* win probabilities: five folds grouped by match, each fold's model calibrated on a disjoint quarter of its own training matches. Every delivery is therefore scored by a model that never saw its match (out-of-fold log loss 0.548). Because folds are random across seasons rather than chronological, a 2015 rating may be produced by a model that has seen 2019 matches. This is appropriate for description but would not be admissible for a forecasting claim.

E. Implementation

Experiments were implemented in Python 3.13 using pandas 3.0, scikit-learn 1.6, XGBoost 3.4, and the SHAP library 0.52. Training was performed on a 16-thread AMD Ryzen 7 PRO 6850U laptop; total wall-clock time for the full pipeline from a cold start is approximately 50 minutes. Code, cached model artefacts and derived rating tables are available at **[repository URL]**.

VI. Results

A. The Instrument

Table III reports held-out performance for the win probability model and both baselines.

TABLE III
WIN PROBABILITY MODEL PERFORMANCE ON HELD-OUT TEST SEASONS

Model	Log loss	Brier	AUC	ECE
Resource-based [9]	0.5638	0.1894	0.794	0.067
Logistic regression	0.5033	0.1696	0.829	0.038
GBM (raw)	0.5201	0.1766	0.815	**0.032**
GBM (isotonic, 1 season)	0.5319	0.1806	0.813	0.054
GBM (isotonic, pooled)	0.5211	0.1766	0.814	**0.032**

95 % CI (match-level bootstrap, $B = 500$) for the pooled-calibration model: log loss $[0.473, 0.577]$, AUC $[0.764, 0.857]$.

Both learned models beat the resource baseline decisively, by roughly 0.05 nats of log loss and 0.02–0.035 AUC. Between them, however, the result is a draw that we report rather than dress up: the regularised logistic regression edges the ensemble on log loss (0.503 versus 0.521) and AUC (0.829 versus 0.814), and the confidence intervals overlap almost entirely.

We read this as the first indication of the paper's theme. Twenty20 win probability is close to a smooth monotone function of a few quantities—projected score against par, wickets in hand, required rate—and with roughly a thousand independent outcomes there is not enough information to identify much beyond that. The binding constraint is the data, not the model class. The ensemble is retained as the instrument for the reasons given in Section IV: it is the best-calibrated specification in the table, and the attribution and SHAP analyses require its interaction structure. Table IV disaggregates its performance.

TABLE IV
TEST LOG LOSS BY INNINGS AND PHASE

Innings	Powerplay	Middle	Death
First	0.6537	0.6446	0.6186
Second	0.5187	0.3776	0.2088

The structure is what the game implies. First-innings loss barely improves from powerplay to death (0.654 to 0.619), because without a target the eventual result remains close to a coin flip however well the innings is read. The second innings is different in kind: loss falls by more than half (0.519 to 0.209) as the chase resolves. Almost all usable information lives in the second innings, which is also where the largest win-probability movements are generated—a fact that returns in Section VI-D.

B. Consequence I: A Silent Model-Selection Failure

TABLE V
EFFECT OF MINIMUM LEAF SIZE ON GENERALISATION

Leaf	Matches	Val. LL	Test LL	Test AUC
30	0.1	**0.4984**	0.6125	0.761
100	0.4	**0.4880**	0.5683	0.782
500	2.1	0.4922	0.5148	0.816
2000	8.3	0.4919	0.5160	0.817
5000	20.8	0.5078	**0.5110**	**0.827**

Leaf size in deliveries; "Matches" divides by the ≈240 deliveries that share one outcome label. All models use `n_estimators` = 400, `max_depth` = 4.

This is the paper's central result. The minimum-leaf hyperparameter is specified in *deliveries*, but the information content of the data is measured in *matches*. A leaf holding 30 deliveries—an unremarkable default, and smaller than several libraries' own—holds roughly one eighth of a single independent observation.

Table V and Fig. 2 show the consequence. As the leaf grows from 30 to 5,000 deliveries, test log loss falls from 0.613 to 0.511 and test AUC rises monotonically from 0.761 to 0.827: a 17 % reduction in loss obtained purely from regularisation, with the curve flattening once leaves exceed roughly two matches' worth of deliveries.

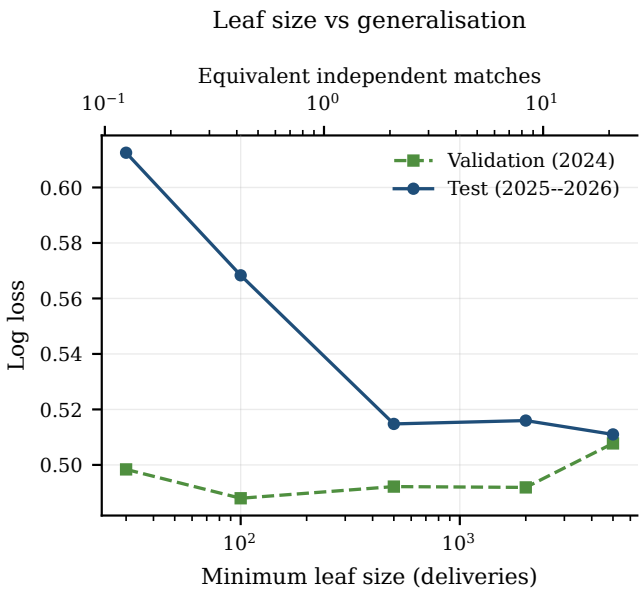


Fig. 2. Minimum leaf size versus generalisation. The secondary axis rescales leaf size into equivalent independent matches. Validation loss is flat over the range across which test loss moves.

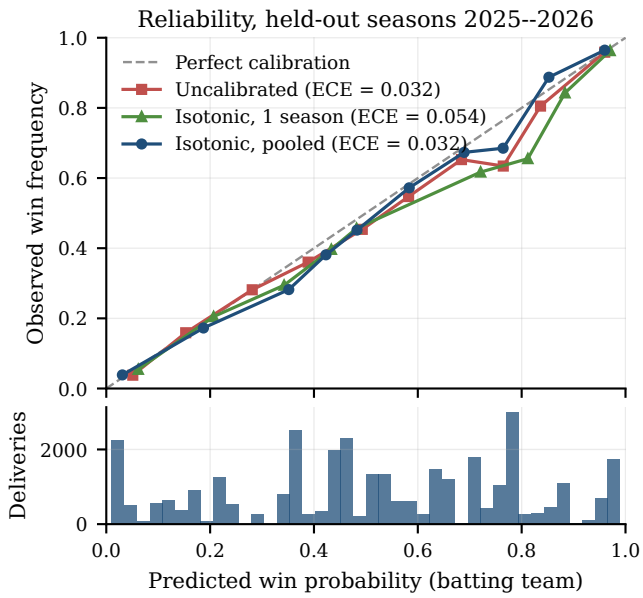


Fig. 3. Reliability diagram for the ensemble uncalibrated, calibrated on one season, and calibrated on pooled out-of-fold predictions, with the density of predictions beneath.

The validation curve, meanwhile, varies by less than 0.02 nats across the entire range, and is *minimised* at the two settings that generalise worst. A practitioner following standard practice—tune on a held-out season, select the lowest validation loss—would therefore choose a model that is 20 % worse on the following two seasons, and would receive no signal warning them of it. This is what we mean by a silent failure: the diagnostic that is supposed to detect over-fitting is itself computed on 71 independent outcomes, and is too noisy and too close to the training regime to see it.

Two recommendations follow. First, regularisation hyperparameters for ball-by-ball models should be specified in independent units—matches, not deliveries—and grids floored accordingly; ours is floored at 500 deliveries for this reason. Second, validation for such models should span multiple seasons or use grouped cross-validation over matches, so that the effective sample size of the selection criterion is not itself a handful of observations.

C. Consequence II: Calibration

The same cause corrupts the post-hoc calibration step. Fitting isotonic regression on the validation season alone—Variant A of Section IV-B, and the textbook recipe—makes calibration *worse*, raising ECE from 0.032 to 0.054 and log loss from 0.520 to 0.532. With 71 independent outcomes the monotone map over-fits, and the corresponding curve in Fig. 3 visibly departs from the diagonal in the upper range. Pooling out-of-fold training predictions with the validation season, Variant B, restores ECE to 0.032.

Two things are worth noting. The uncalibrated ensemble tracks the diagonal closely to begin with, with a mild optimism in the 0.6–0.85 band; its ECE of 0.032 is the lowest of the five specifications in Table III, better than the logistic baseline's 0.038 despite that model's superior log loss. And post-hoc calibration is therefore neutral at best here: the honest summary is that it cannot help a model that is already calibrated, and that fitted on too few independent outcomes it will actively hurt.

D. Consequence III: Rating Stability

TABLE VI
SEASON-OVER-SEASON SPEARMAN RANK CORRELATION OF PLAYER RATINGS

Metric	Batters	Bowlers
Batting average / Bowling average	0.193	0.106
Strike rate / Economy rate	**0.413**	**0.480**
WPA	0.143	0.144
WPA_{100}	0.133	0.107

$n = 560$ batter and $n = 534$ bowler consecutive-season pairs, qualification threshold 120 deliveries.

The third consequence appears once the model's outputs are aggregated into a player rating. WPA is markedly *less* stable season-over-season than the conventional rate statistics: strike rate reaches a rank correlation of 0.413 for batters against 0.133 for WPA_{100}, roughly three times as stable, and for bowlers economy rate reaches 0.480 against 0.107, a factor of four and a half. WPA is comparable to the traditional *averages*, which are themselves weakly persistent.

Two mechanisms operate here and they must be kept distinct, because they have opposite implications for whether the problem is fixable.

The first is effective sample size again, now at player level. A qualification threshold of 120 deliveries does not deliver 120

observations' worth of information, because win-probability movement is heavily concentrated: second-innings death overs are 10.0 % of all deliveries but carry 17.9 % of total $|\Delta|$ mass, a mean of 0.027 win-probability points per ball against 0.015 across the corpus. A player's seasonal rating is dominated by a handful of decisive events, so its sampling variance is far larger than the delivery count suggests. This component is a statistical problem, and hierarchical shrinkage toward role-specific priors would mitigate it.

The second is not a statistical artefact at all. Whether a player is on strike or holding the ball at those decisive moments depends on batting order, match situation and captaincy—circumstance, not skill—and circumstance does not persist across seasons the way an ability does. No amount of shrinkage recovers a signal that is not there.

The practical consequence is the same under either mechanism, and it is a real limitation on what this class of metric can be used for. WPA remains a defensible *descriptive* account of what a player contributed to matches already played. It is not, on this evidence, a superior basis for *projecting* future contribution, which is what retention and auction decisions actually require.

E. Consequence IV: Validation Power

TABLE VII
NEXT-SEASON MATCH OUTCOME PREDICTION FROM AGGREGATED SQUAD RATINGS

Rating basis	AUC [95% CI]	Log loss
Conventional metrics	0.528 [0.437, 0.616]	0.6929
WPA	0.540 [0.440, 0.640]	0.6937
Combined	0.524 [0.433, 0.609]	0.6931

Evaluated on 140 test-season matches.

The fourth consequence concerns the design of validation experiments rather than of models. No rating basis separates next-season match outcomes better than chance: every interval in Table VII comfortably contains 0.5, and the three specifications are indistinguishable. WPA's nominal 0.540 is not a positive finding.

The correct reading is underpowered rather than disproved, and the arithmetic is the point. A test season supplies 140 independent matches; the resulting bootstrap half-width is approximately 0.09 AUC. Any true effect of prior-season squad quality on a single Twenty20 match is plausibly well inside that band—IPL squads are deliberately equalised by auction and retention rules, and one match is close to a coin flip regardless. The experiment as designed could not have detected the effect it was built to detect, and reporting the null without that arithmetic would invite the wrong conclusion. Resolving the question needs more independent matches: pooling several leagues, or a rolling-origin design over all nineteen seasons rather than two.

F. Model Interpretability

Having reported four ways the instrument can mislead, we check that it is measuring the right thing. Fig. 4 ranks

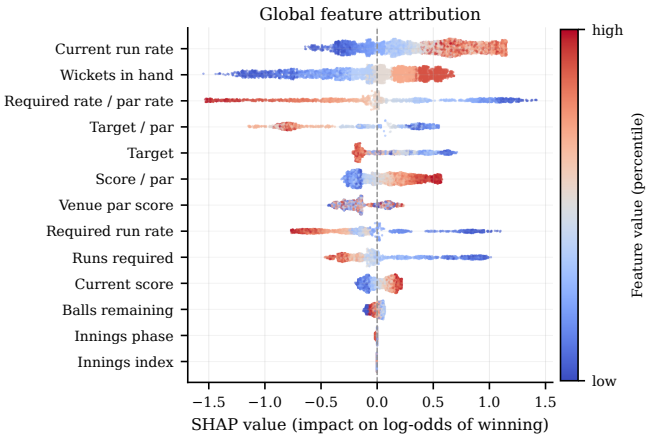


Fig. 4. Global feature attribution for the win probability model, features ordered by mean absolute SHAP value.

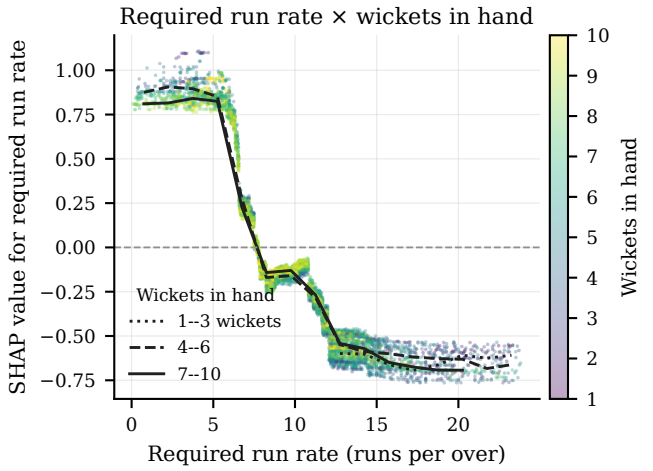


Fig. 5. Interaction between required run rate and wickets in hand, with median trends by wicket band.

the drivers of win probability by mean absolute SHAP value: current run rate (0.496), wickets in hand (0.363), par-relative required run rate (0.323) and par-relative target (0.307) dominate, followed by the absolute target (0.212) and par-relative score (0.206). The directions are as cricketing intuition requires—high current run rate and abundant wickets push the log-odds up, high required rates push them down. Two features are near-inert: innings phase (0.004) and the innings index itself (0.000). The latter is not a defect but a consequence of the encoding: the missingness pattern of the target and required-rate features already identifies the innings perfectly.

The three par-relative features together account for more attributed mass than their absolute counterparts (0.836 against 0.505), confirming that the era normalisation of Section III does real work.

Fig. 5 plots the interaction the attribution logic most depends on. The required-run-rate contribution falls steeply and monotonically, crossing zero at approximately 8 runs per over—

close to the par scoring rate—and flattening beyond about 13, by which point the chase is effectively decided. The wicket bands separate at the high end, but in the opposite direction to the intuition usually offered: at required rates above 13, teams with 7–10 wickets in hand receive a *more* negative contribution from the required rate than teams with 1–3. This is not a paradox. When few wickets remain, the wickets feature has already absorbed most of the bad news, leaving the required rate less to explain. The joint effect on win probability is monotone in the sensible direction; it is the *decomposition* that shifts.

G. Illustrative Case Study

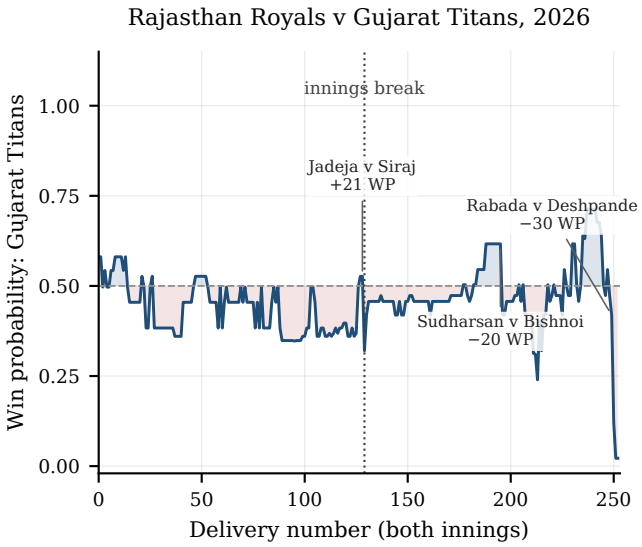


Fig. 6. Ball-by-ball win probability trace for Rajasthan Royals versus Gujarat Titans, Ahmedabad, 2026, a held-out test match.

Fig. 6 traces a close held-out match. Gujarat Titans' win probability oscillates around even money for almost the entire contest, crossing 0.5 repeatedly through both innings before collapsing to 0.02 on the final delivery.

Three deliveries carry the largest swings. Jadeja's strike against Siraj at the end of the first innings is worth $+20.7$ win-probability points because it lands at the innings boundary where the target is finally fixed—a contribution the scorecard records as a handful of runs among two hundred. Sai Sudharsan's dismissal by Bishnoi costs 19.8 points, and Rabada's over to Deshpande in the closing stages swings 29.8 points back against the chase. The allocation is instructive precisely where it departs from the scorecard: a late over conceding few runs but arriving with the match in the balance outranks an economical spell bowled while the result was never in doubt. Economy rate cannot express that distinction; a win-probability metric prices it directly.

H. Divergence from Conventional Metrics

That the two families of metric disagree is not in doubt: among qualified 2026 batters the largest rank shifts exceed thirty places in a field of roughly sixty. Table VIII shows the pattern is dominated by high strike-rate players whom WPA demotes, while Jadeja, a middle-order accumulator ranked 49th on strike rate, rises to 14th on impact. The bowling side shows clearer role logic: Deshpande rises 35 places and Rabada 28 relative to their economy ranks, both death-overs specialists whose economy is inflated by the phase they operate in, while spinners Suyash Sharma and Shami fall 34 and 30 places from economy ranks earned largely in the middle overs.

TABLE VIII
LARGEST RANK DIVERGENCE BETWEEN WPA AND STRIKE RATE, SEASON 2026

Player	Conv. rank	WPA rank	Δ
KL Rahul	11	50	−39
RA Jadeja	49	14	+35
TM Head	14	47	−33
D Ferreira	10	40	−30
FH Allen	2	32	−30

Positive Δ: WPA ranks the player higher than strike rate does. Qualification threshold 120 deliveries faced.

Two cautions are required, both following from Section VI-D. A single season's ranking is itself a high-variance quantity, so individual rows should be read as illustrative rather than as findings about those players. And a divergence between WPA and conventional statistics is not evidence of mispricing—it may equally reflect WPA's higher variance. We therefore claim only that the two families rank players very differently, and in a manner that tracks role, and we leave valuation to work that can join ratings to prices.

I. What Aggregation Buys, and Where It Stops

TABLE IX
CHENNAI SUPER KINGS HEAD-TO-HEAD RECORD AND MEAN PER-MATCH WPA DECOMPOSITION

Opp.	M	Win %	Bat WPA	Bowl − Bat
MI	41	48.8 [34, 63]	-0.046	-0.066 to +0.242
RCB	35	60.0 [43, 77]	-0.007	-0.093 to +0.282
DC	33	63.6 [48, 79]	+0.047	-0.147 to +0.168
RR	32	50.0 [31, 69]	-0.019	-0.099 to +0.204
PBKS	31	48.4 [31, 65]	+0.028	-0.263 to +0.118
KKR	30	66.7 [50, 83]	+0.076	-0.139 to +0.145
SRH	24	62.5 [42, 79]	+0.068	-0.196 to +0.149
All	259	56.0 [50, 62]	+0.011	-0.026 to +0.084

Brackets are 95% match-level bootstrap intervals ($B = 500$). Opponents with fewer than 20 fixtures are omitted as underpowered. Per-match WPA is denominated in win probability, so a team's total is close to $+0.5$ when it wins and -0.5 when it loses; the batting/bowling split is the informative quantity, and the final column gives the interval for that difference.

The natural response to everything above is to aggregate until the effective sample size is adequate. This subsection asks how far that gets, using the most favourable case available: Chennai Super Kings, the most-capped franchise in the competition, with 259 completed fixtures across nineteen seasons.

Two things are genuinely gained. The decomposition is *exact*: by the identity of Section IV a team's batting and bowling

contributions sum to its result, so the question "was this record built with bat or ball?" is well posed in a way no combination of strike rate and economy rate makes it. And the aggregate quantities are, marginally, resolvable. CSK's overall win rate is 56.0 % with a 95 % interval of $[49.8, 62.0]$, and their mean per-match bowling WPA of $+0.042$ $[+0.002, +0.085]$ excludes zero while their batting WPA of $+0.011$ $[-0.031, +0.053]$ does not.

It is exactly here that the analysis must stop, and the reason is instructive. That bowling WPA is distinguishable from zero while batting WPA is not does *not* license the conclusion that bowling contributes more: that is the difference-of-significance fallacy. The quantity the claim requires is the difference itself, and it is $+0.030$ with a 95 % interval of $[-0.026, +0.084]$. The interval contains zero. Over 259 matches and nineteen seasons, the most-capped franchise in the league does not supply enough independent observations to establish which discipline wins its matches. We had drafted the opposite claim before computing the interval, and record that here because the error is an easy one to make.

The per-opponent rows make the same point more starkly. Table IX and Fig. 7 show that every individual rivalry interval straddles the null: no head-to-head win rate is distinguishable from even, and no batting-versus-bowling difference is distinguishable from zero. Five further opponents are omitted entirely with fewer than 20 fixtures. The point estimates are suggestive—CSK's weakest record is against Mumbai Indians and their strongest against Kolkata Knight Riders—but they are point estimates over thirty-odd matches, and the intervals say so.

TABLE X
LEADING CAREER WPA CONTRIBUTORS FOR CHENNAI SUPER KINGS

Player	Deliveries	Career WPA
R Ashwin	2,373	+2.13
DJ Bravo	3,050	+1.83
M Muralitharan	888	+1.82
MM Sharma	1,020	+1.57
MS Dhoni	3,354	+1.49
RA Jadeja	4,720	+1.44

Deliveries faced plus deliveries bowled while representing the franchise; minimum 240. Career totals aggregate over far more deliveries than a seasonal rating, but remain subject to the leverage concentration of Section VI-D.

Table X lists the franchise's leading career contributors, and illustrates what the metric does well when it is used descriptively. Five of the six are bowlers or bowling all-rounders. Muralitharan is the sharpest illustration of leverage weighting: he accumulates the third-highest career total from 888 deliveries, under a third of Jadeja's 4,720, because his contributions arrived disproportionately at decisive moments. No conventional aggregate ranks a player above teammates with five times the workload. We present this as a description of what occurred, not as an estimate of ability.

VII. DISCUSSION

A. Implications for Practice

The recommendations that follow are cheap to adopt and, on this evidence, consequential.

Count independent outcomes before choosing hyperparameters. Regularisation expressed in rows is meaningless when rows are not observations. Divide the nominal count by the cluster size and grid accordingly; in our case a floor of 500 deliveries per leaf, about two matches, was the difference between 0.61 and 0.51 test log loss.

Do not trust a single held-out season. Validation loss over 71 independent outcomes was flat across a range where test loss moved by a fifth, and pointed the wrong way. Use several seasons, or grouped cross-validation over matches, and treat validation loss differences smaller than the noise floor as absent.

Calibrate on enough independent outcomes, or not at all. A post-hoc calibrator is a fitted model and obeys the same arithmetic. Ours degraded a well-calibrated ensemble when fitted on one season and was neutral when fitted on a thousand outcomes.

Report intervals on descriptive statistics too. The head-to-head analysis of Section VI-I would have carried an attractive and false headline had we not bootstrapped it. Descriptive work is not exempt.

For player evaluation specifically, the position we end at is narrower than the one this class of metric is usually asked to occupy. A win-probability rating is a good descriptive instrument: it prices each delivery by its effect on the quantity teams care about, it decomposes exactly at team level, and it recovers role structure that economy rate and strike rate obscure. It is not, on this evidence, a better forward-looking valuation input than the conventional aggregates, whose stability—whatever their context blindness—is their genuine virtue.

B. Limitations

Single competition. Every result here comes from one league. The mechanism we describe is arithmetic rather than cricketing, and we expect it wherever labels are shared within a contest, but that expectation is an argument rather than a demonstration. Replication on other Twenty20 competitions, and on other sports with possession- or pitch-level data, is the obvious next test.

Attribution granularity. Run-outs and fielding contributions are not attributed to the responsible fielder, so defensive value is omitted from player ratings.

Extras convention. Wides and no-balls are credited wholly against the bowler with no offsetting batter credit. This is defensible for wides but questionable for no-balls, which the batter does face; the convention depresses batting WPA slightly at player level, and is why Section VI-I uses the raw increment instead.

Opponent quality. The state representation excludes team and player identity, so a rating absorbs the quality of opposition faced. The head-to-head decomposition inherits this.

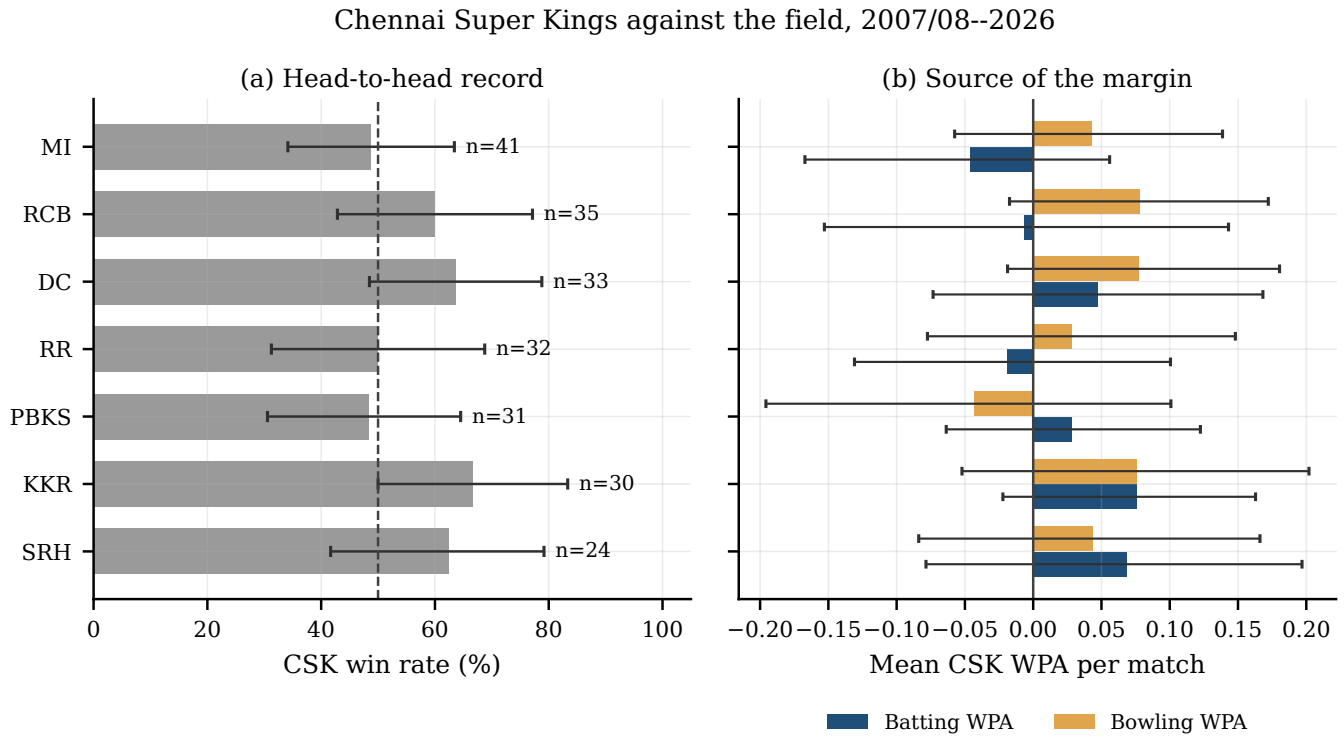


Fig. 7. Chennai Super Kings against the field, 2007/08–2026, opponents with at least 20 fixtures. (a) Head-to-head win rate with 95 % match-level bootstrap intervals; every interval straddles 50 %. (b) The same records decomposed into mean per-match batting and bowling WPA with the same intervals. The decomposition is exact, but 259 matches are not enough to resolve it.

Non-striker and partnership effects. Credit accrues only to the striker, so a non-striker who rotates strike or absorbs pressure receives none.

Regime shift. The par-relative features substantially mitigate two decades of scoring inflation but do not eliminate it; the temporal split measures the residual rather than removing it.

Rating basis. Player ratings use out-of-fold probabilities whose folds are random across seasons, which is appropriate for description but inadmissible for forecasting.

C. Future Work

The most direct extension is more independent matches: pooling several Twenty20 competitions would address the power limitation of Section VI-I directly, and would test whether the model-selection failure reproduces at larger effective sample sizes or is specific to this regime. Hierarchical shrinkage of player ratings toward role-specific priors would mitigate the sampling component of the stability problem, though not the circumstantial component, and the gap between those two is itself worth measuring. Fielder-level attribution and opponent adjustment would improve the instrument. Finally, the team-level decomposition generalises immediately to a full league-wide matrix, of which Section VI-I presents one row—with the caveat that, on the evidence here, most cells of that matrix will not be resolvable either.

VIII. Conclusion

Ball-by-ball sports data invites a mistake that its file format makes almost automatic: treating the delivery as the unit of observation. In the Indian Premier League record, 286,621 deliveries carry 1,195 independent outcomes, and the factor of 240 between those numbers propagates into every stage of a modelling pipeline.

We documented four consequences using a calibrated win-probability model as the instrument. Model selection fails silently: an unregularised ensemble records the best validation loss and generalises 20 % worse, with validation loss flat across the range over which test loss moves. Post-hoc calibration fitted on a single season degrades a well-calibrated model, raising expected calibration error from 0.032 to 0.054. A leverage-concentrated rating is three to four times less persistent season-over-season than strike rate, because 10.0 % of deliveries carry 17.9 % of all win-probability movement. And a validation design with 140 test matches cannot resolve the effects it was built to detect.

Aggregation helps but does not rescue. The team-level decomposition is exact by construction, and for the most-capped franchise in the league the aggregate margin is marginally resolvable—yet 259 matches across nineteen seasons still cannot establish whether that margin is built with bat or ball, the difference being $+0.030$ with an interval of $[-0.026, +0.084]$.

None of this argues against win-probability methods, which

remain the right way to price a delivery by what it does to the result, and which recover role structure that aggregate statistics cannot express. It argues that the number of rows in a ball-by-ball file is not the number of things one has observed, and that regularisation, validation, calibration and reporting should all be specified in the units that actually carry information.

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